Humidification

Final Lab Report

Unit Operations Lab 1

February 6, 2020

Group #2, Thursday, AM section

Diana Awimrin

Leena Daniel

Jay Rawal

Esteban Vazquez

Commented [DL1]: 87/100

1. Executive Summary

The objective of the experiment is to study the fundamental concept of heat and mass transfer in a wetted wall column by recording humidity and temperature profiles. With the information in the experiment, it can be compared with existing correlations to test the accuracy of the data and obtain the heat and mass transfer coefficients. The humidification unit is a 4 ft. (1.22 m) long by 2 in. (0.05 m) diameter cylindrical plexiglass-column with introduction of either cold or hot air flow from the bottom with varying flows of water trickling down top to bottom. For humidification to occur, varying temperatures were used in the experiment from ambient air temperature of 27 °C and two warmer temperature of 45 °C and 65 °C. The flow of water was also varied from three values: 10.19, 14.44, and 18.69 m^3/hr. (6, 8.5, and 11 SCFM respectively) in order to obtain enough data to calculate trends and compare them to theoretical sections.

The mass transfer coefficient from the experiment in *Figure 1* shows the positive trend with relation to Reynold's number increasing along with the mass transfer coefficient. This confirms the theoretical data as increasing the flow of water will in turn increase Reynold's number and thereby increasing mass transfer coefficient.

The heat transfer coefficient, h, from the experiment in *Figure 2* shows the increasing trend of heat transfer coefficient with Reynold's number. However, one of the interesting results were that no heat introduced in the system allowed for a higher value of heat transfer coefficient whereas the introduction of heat translated to lower maximum value for h. There could be a mixed correlation between heat and mass transfer coefficients. This will be further explained in the Discussion section as to why there is some sort of discrepancy. Additionally, Appendix II will further go over the error analysis where the experiment could have had some issues with accurate temperature and humidity gradient at wet bulb thermocouple.

Commented [DL2]: 8/10

Commented [DL3]: Water flow rate or air flow rate? Anyway, what's the other component's flow rate?

Commented [DL4]: Better to use SI units.

Commented [DL5]: Good, but not enough specific detail to reinforce your conclusion. What are the other trends in this experiment? Like the humidity, NtG and other dimensionless groups.

Commented [DL6]: What's the trends between the coefficients and the temperatures?

2. Introduction

The objective of this experiment is to examine mass and heat transfer in a wetted wall column and acquire experimental values for the heat and mass transfer coefficients. These values will be obtained through experimental methods and will be compared to values calculated using the theoretical equations. The system variables that will serve as the independent variables for this experiment include liquid flowrate, air flowrate, and temperature of the incoming air. The scope of this lab is to understand that when there is mass transfer, there must also be heat transfer. In this lab, it is understood that when there is a component that transfers from the liquid phase to the gas phase, heat is associated within the process. This process can be proven in real-world applications such as distillation columns, absorption, and boilers.[1] Engineers can explain that liquid-liquid extraction takes place in a distillation column in the industries involving alcohol and gas. Engineers must also account for the heat transfer rate which can slow down or speed up the rate at which mass is transferred. There are applications in which mass and heat transfer occur simultaneous and this can be shown through evaporation and condensation.[2] In this lab, the apparatus of humidification is utilized in order to see the effects of heat and mass transfer using only air and water. Also, the experimental values were obtained for both mass and heat transfer where the coefficients will be plotted as a function for the Reynolds numbers and compared to the theory explained below. The process of the apparatus is as follows: water is pumped in the top of column through a water reservoir. The flow will have to be increased in order for the water to flow through all parts of the column. In the first part of the experiment temperature will be recorded at seven different locations as well as the percent of humidity. This was done for water flow without heat, water flow with heat, and heat with no water flow.

Commented [DL7]: 9/10
Incomplete section.

Commented [DL8]: Good.

Commented [DL9]: Good.

Commented [DL10]: Good. What’s the next part?

3. Theory

Humidification involves the transfer of material between a pure liquid phase and a fixed gas that is nearly insoluble in water. The fixed gas in this experiment will be natural air. The gas phase and the liquid phase will influence each other, and their respected connections will change simultaneously. The operations performed in lab and the real-world application are proven to be simpler than those of an absorber or a stripper. This is only true when the system contains one component. To follow with the theory from the book, it is also noticed that there is no concentration gradient and no resistance to mass transfer or liquid phase.[3] It is also important to note, that there is going to be heat transfer and gas phase mass transfer within the column. These two properties would influence each other but should be treated and studied separately. The engineering basis followed and calculated for was vapor-free gas. Vapor was present at the top of the apparatus. Vapor is the gaseous form of the component that is present in the liquid and gas phase. The properties of gas-vapor mixture can vary, as shown in the lab, therefore, the total pressure is kept constant. Gas-vapor mixture can vary due to total pressure. That is why during the experiment we made sure to keep total pressure at a constant when recording the vapor. From the book it was stated that it can be assumed that the gas and vapor phase will follow the rule of 1 atm. The factor of humidity was also noted, which is the vapor carried by a unit of mass of vapor-free gas. Humidity was noted in the lab when the heat exchanger was turned on and played in the factor of air being released from the system. Humidity depends only on the partial pressure of the partial pressure of the vapor in the mixture. This is only true when the pressure is constant. The humidity of the apparatus is calculated from the equation below. [1]

$$\boldsymbol{\mathcal{H}} = \frac{M_A p_A}{M_B(P - p_a)} \qquad (1)$$

Where, M_A and M_B are the molecular weights of the components of A and B.

Then, the humidity is related to the mole fraction in the gas phase by the following equations shown below.

$$y = \frac{\mathcal{H}/M_A}{1/M_B + \mathcal{H}/M_A} \qquad (2)$$

$$\ln\left(\frac{y_i' - y_1'}{y_i' - y_z'}\right) = \frac{k_y P_z}{G_s} \qquad (3)$$

Commented [DL11]: 7/10
Good introduction section, but equations like Fick's law, Newton's law, Dimensionless groups such as Re, Sc, Sh, transfer coefficient Kc and h, NtG and their assumptions are needed.

Commented [DL12]: References.

Commented [DL13]: Units.

$$\ln\left(\frac{T_i - T_1}{T_i - T_2}\right) = \frac{hP_z}{G_s C_s} \qquad (4)$$

Humidity problems can be solved independently through the equations above. Y1 and T1 are the inlet composition and temperature and Y2 and T2 are the outlet composition and temperature. The left side of the first equation is related to the number of times the average driving force for mass transfer is divided into the difference in composition between the top column and bottom of the column.

4. Results

Commented [DL14]: 15/15

In this experiment, the objective was to study the transport of both mass and heat in a wetted wall column and to obtain experimental values for the heat and mass transfer coefficients. Water is filled up in the reservoir tank and three types of experiments were done. In the first measurement, the control was taken with the water flowing through the column. The humidity was measured near the wet bulb thermocouple. Along with measuring the humidity, different temperatures were measured at six different thermocouples: Wet Bulb, Dry Bulb, Top Column, Middle Column, Bottom Column, and the Reservoir Column. These measurements were taken in different conditions; there was a control taken with the water flowing but with no steam or cold air coming in. Then, experiments using three different cold air flowrates were done with the following conditions: cold air with no water flowing, cold air with water flowing, and hot air with temperatures at 45°C and 65°C with water flowing. The data from this experiment can be seen in Table 1, in Appendix 1.

Commented [DL15]: Better to specify the temperature used.

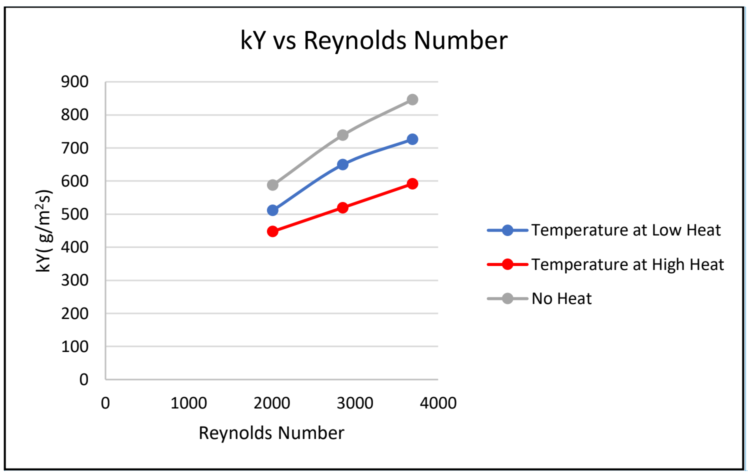

Figure 1: Plot of k_Y as a function of Reynolds number

In figure 1, above, the plot shows the mass transfer coefficient, k_Y, as a function of Reynolds number. It is seen that the k_Y is directly proportional to the Reynolds number as the slope is moving in the positive direction. This was the expected result due to the relationship Reynolds number has with the Sherwood number. Through the following equations 3 through 5, below, the relationship between Reynolds number and k_Y can be seen.[4] If the Reynolds number increases, the Sherwood number would increase and that would cause the mass transfer coefficient k_c to increase. If k_c increases, then k_Y certainly will increase, which in summary is shown through figure 1.

Commented [DL16]: Good discussion.

$$Sh = 0.023 \cdot Re^{0.83} \cdot Sc^{0.44} \quad (3)$$

$$Sh = k_c \cdot \frac{L}{D_{AB}} \quad (4)$$

$$k_Y = k_c \cdot (MW_{AIR}) \cdot c \cdot (1 - y)_M \quad (5)$$

Also, since the mass transfer coefficients aren't affected directly by the temperatures, it can be seen that as the temperature was increased, the values for the mass transfer coefficients decreased.

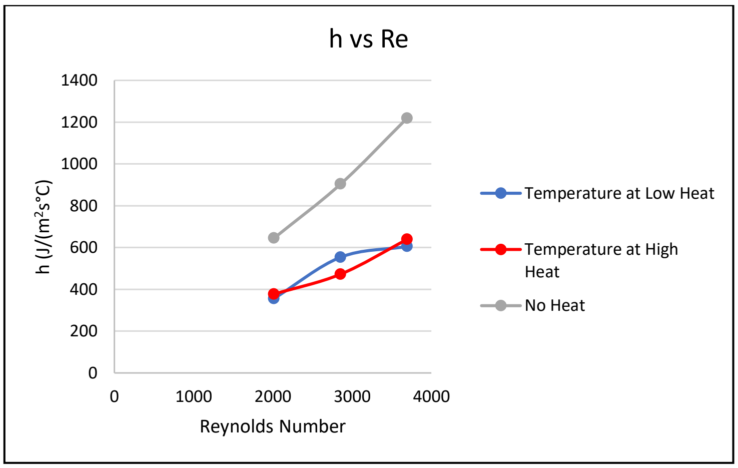

Figure 2: Plot of h as a function Reynolds number

In figure 2, above, the plot shows the heat transfer coefficient, h, as a function of Reynolds number. It can be seen that the heat transfer coefficient is high when there is no heat and it somewhat looks like it decreases with higher heat. Even though the temperatures are affecting the heat transfer coefficient, the relationship between the coefficient and the Reynolds number is still proportional to each other. The reason for this could be due to the heat transfer coefficient being influenced by the mass transfer coefficient. This relationship causes the relationship between the heat transfer coefficient and the Reynolds number to go in a positive direction.

Commented [DL17]: Good discussion.

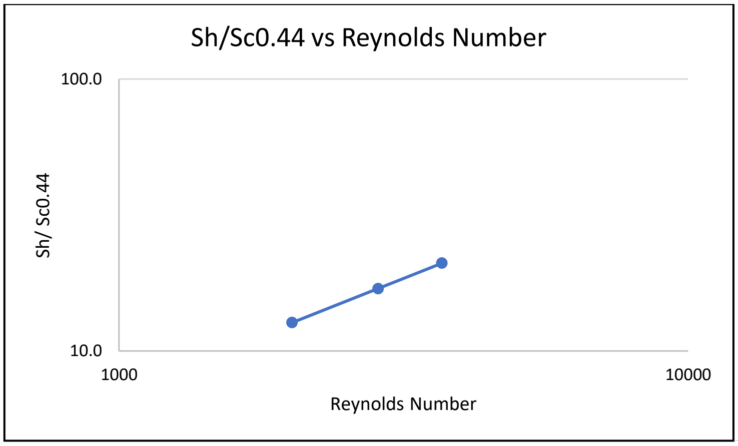

Figure 3: Plot of the Sherwood number divided by the Schmidt Number to the 0.44 as a function of Reynolds number

Commented [DL18]: Mind the upper characters.

In figure 3, above, the plot of the Sherwood number divided by the Schmidt number as a function of the Reynolds number is shown in a $\log_{10}$-$\log_{10}$ plot. In this plot, the relationship of the Sherwood number and Schmidt number with the Reynolds number, shown in equation 3, can be seen here. The plot shows the slope being a positive slope, which indicates that they are both directly proportional to each other. This occurs because the Schmidt number is divided on both sides of the equation, what occurs to the Sherwood number will occur to the Reynolds number. In figure 4, below, the NTU is plotted as a function of the inlet air temperature with the air flowrate as a parameter. As the temperature increases, the amount of the humidity decreases and while the flow rate is increasing.

Commented [DL19]: Good.

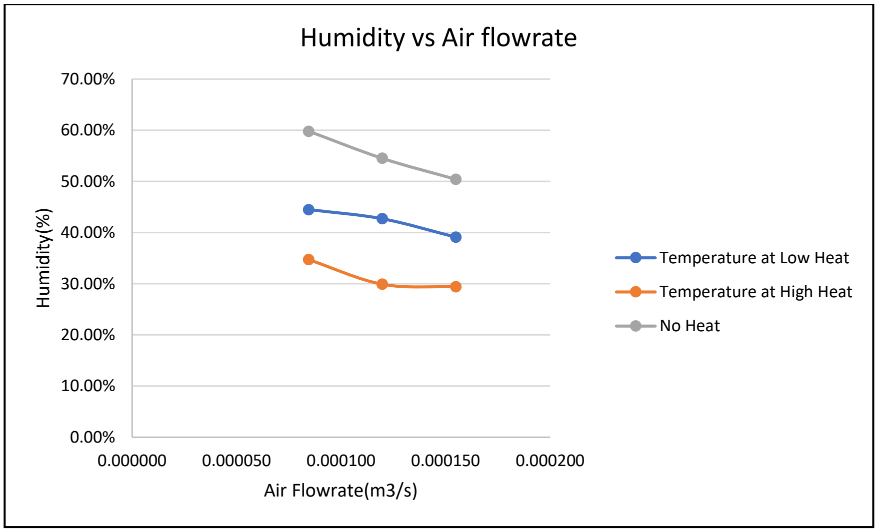

Figure 4: NTU as a function of inlet air temperature with the air flowrate as a parameter

5. Discussion

Commented [DL20]: 13/15
Why the RH% changes with parameters change? Discussions are needed.

As previously mentioned, this experiment is very important in terms of understanding the transfer trends between mass and heat in equipment used in industry. The humidification unit set up for this experiment allowed for the analysis of correlating heat transfer and mass transfer with the use of water and air flow. The data collected was used to calculate the mass and heat transfer coefficients which are then compared to theoretical values.

To begin, it can be noted that as the mass transfer coefficient increases, the Reynolds number also increases as seen in Table 1. This trend is seen among all the different trials performed: low heat, high heat and no heat. These results follow the same conclusion stated in the theory. The reason for the proportionality between these two variables relates to the Sherwood number. When the Sherwood number increases, the mass transfer coefficient is also expected to increase. Since the system we use does have heat energy, there is influence of heat transfer coefficient and mass transfer coefficient resulting in Sherwood number being affected by mass transfer coefficient. If the system was non-isothermal, then there will be no correlation between mass and heat transfer as it indicates the system is not influenced by heat diffusion, which is not the case. Next, the heat transfer coefficient is analyzed. In relation to mass transfer, the heat transfer sees similar trends. As the heat transfer is increased, the Reynolds is also increased. This follows literature because just as before, these coefficients are proportional to each other.

Commented [DL21]: The reason and the result are reversed.

Commented [DL22]: Good.

It is also important to note that as the heat slowly decreased from high to low and then to none, the mass transfer increased. This was not expected because as mass transfer increased the heat should also increase. The reason for this discrepancy is because after the trial with heat was completed, the membrane around the wet bulb thermocouple fell off causing the experimental values for the rest of the experiments to be off by a certain factor. This will affect the value of humidity we will generate which in turn will make the humidification experiment giving us a factorable error from real world scenarios. This could have been eliminated by fixing the membrane and recording the values correctly but due to time constraints, the experiment needed to continue. To completely eliminate the error, a specialized thermocouple for wet bulb can be added that does not allow manual interaction in order to read the value. Having the least disturbances on the system will allow for a more controlled result.

Commented [DL23]: Good thought.

In conclusion, the experiment gave us results that were expected from the experiment. The values of Reynold's number coincided with heat and mass transfer coefficients. The questions that remained unanswered in the experiment is to confirm the trend of no heat, low heat, and high heat values with humidification ratio. This was in part due to the membrane falling off the thermocouple which will give us a factor value that may be off from the low and high heating values.

Commented [DL24]: How the experimental data correlates to the theoretical data? If there're some differences, give a discussion.

6. References

[1] Fog, Smart. “3 Main Uses for Industrial Humidification Systems.” Smart Fog, 23 Jan. 2020, www.smartfog.com/3-main-uses-industrial-humidification-systems.html.

[2] “How Columns Work?” *SRS Engineering Home*, www.srsengineering.com/our-products/distillation-columns/how-columns-work/.

[3] Smith, Julian C, and Peter Harriott. “19: Humidification.” *Unit Operations of Chemical Engineering*, by Warren L McCabe, Seventh ed., McGraw-Hill International Editions, 2005.

[4] Zdunek, A.CHE 381:Unit Operations Laboratory-Humidification, Spring 2018, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL.

[5] Final Report Template Fall 2009, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL.

[6] CHE4162 Syllabus Spring 2010, Department of Chemical Engineering, Louisiana State University, Baton Rouge, LA.

Commented [DL25]: 7/10
[5] and [6] are not found in the report.

Commented [DL26]: Date of access.

Commented [DL27]: Pages.

7. Appendix I: Data Tabulation/Graphs

Commented [DL28]: 10/10

Table 1: Collected Data for the Control and the Cold Air

	Cold Air					
	No Water Flowing			Water Flowing		
	Cold Air Flowrate (SCFM)	Cold Air Flowrate (SCFM)	Cold Air Flowrate (SCFM)	Cold Air Flowrate (SCFM)	Cold Air Flowrate (SCFM)	Cold Air Flowrate (SCFM)
	6	8.5	11	6	8.5	11
Thermocouple Reading	14°C	16°C	20°C	20°C	22°C	20°C
T1: Wetted Bulb(Hydrometer)	23.1°C	23.2°C	24.8°C	23.8°C	24.2°C	24.2°C
T2: Dry Bulb	23°C	23°C	23°C	24°C	25°C	24°C
T3: Top Column	22°C	22°C	21°C	27°C	28°C	27°C
T4:Mid Column	23°C	23°C	23°C	26°C	27°C	27°C
T5:Bottom Column	23°C	23°C	23°C	26°C	26°C	26°C
T6: Reservoir Column	26°C	26°C	26°C	25°C	25°C	25°C
Humidity	6.10%	7.20%	8.40%	59.80%	54.50%	50.40%

Table 2: Collected Data for the Heated Air Experiments

	Hot Air					
	Temperature at 45°C			Temperature at 65°C		
	Cold Air Flowrate (SCFM)	Cold Air Flowrate (SCFM)	Cold Air Flowrate (SCFM)	Cold Air Flowrate (SCFM)	Cold Air Flowrate (SCFM)	Cold Air Flowrate (SCFM)
	6	8.5	11	6	8.5	11
Thermocouple Reading	26°C	25°C	25°C	25°C	24°C	25°C
T1: Wetted Bulb(Hydrometer)	35.5°C	34.9°C	35.3°C	38.4°C	39.7°C	38.6°C
T2: Dry Bulb	40°C	40°C	40°C	46°C	48°C	49°C
T3: Top Column	34°C	33°C	33°C	31°C	31°C	30°C
T4:Mid Column	32°C	32°C	32°C	30°C	30°C	29°C
T5:Bottom Column	32°C	32°C	31°C	30°C	30°C	29°C
T6: Reservoir Column	31°C	31°C	30°C	30°C	29°C	29°C
Humidity	44.50%	42.70%	39.10%	34.70%	29.90%	29.40%

Table 3: Calculated Results

Heated Temperature 45°C			Heated Temperature 65°C			No Heat		
Air FlowRate (LPM)								
5.07	7.19	9.30	5.07	7.19	9.30	5.073	7.18675	9.3005
Airflowrate(m^3/s)								
8.46E-05	1.20E-04	1.55E-04	8.46E-05	1.20E-04	1.55E-04	8.46E-05	1.20E-04	1.55E-04
G's (g/s)								
1.08E+02	1.53E+02	1.98E+02	1.08E+02	1.53E+02	1.98E+02	1.08E+02	1.53E+02	1.98E+02
y2								
2.76E-03	2.65E-03	2.42E-03	2.15E-03	1.85E-03	1.82E-03	3.70E-03	3.37E-03	3.12E-03
Y'2								
1.72E-03	1.65E-03	1.51E-03	1.34E-03	1.15E-03	1.13E-03	2.31E-03	2.10E-03	1.94E-03
y1								
3.79E-04	4.47E-04	5.22E-04	3.79E-04	4.47E-04	5.22E-04	3.79E-04	4.47E-04	5.22E-04
Y'1								
2.35E-04	2.78E-04	3.24E-04	2.35E-04	2.78E-04	3.24E-04	2.35E-04	2.78E-04	3.24E-04
Interface Humidity								
50.60%	49.90%	47.50%	40.80%	37.10%	37.80%	65.90%	61.70%	58.80%
yi								
3.13E-03	3.09E-03	2.94E-03	2.53E-03	2.30E-03	2.34E-03	4.08E-03	3.82E-03	3.64E-03
Y'i								
1.95E-03	1.93E-03	1.83E-03	1.57E-03	1.43E-03	1.46E-03	2.54E-03	2.38E-03	2.27E-03
ln(yi-y1)/(yi-y2)								
1.99	1.78	1.54	1.74	1.42	1.25	2.28	2.02	1.79
ky								
512	650	727	448	520	592	588	739	846

Table 4: Continued Results Calculated

Reynolds Number								
2015	2855	3695	2015	2855	3695	2015	2855	3695
ln((Ti-T1)/(Ti-T2))								
1.37	1.50	1.27	1.45	1.28	1.34	2.47	2.44	2.55
Cs = Cair + Y' Cwater								
1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01
h								
356	554	605	377	473	639	645	904	1219
schmidt number								
0.0072	0.0072	0.0072	0.0072	0.0072	0.0072	0.0072	0.0072	0.0072
Sherwood Number								
1.45	1.93	2.39	1.45	1.93	2.39	1.45	1.93	2.39
Sh/Sc^0.44								
12.7	17.0	21.0	12.7	17.0	21.0	12.7	17.0	21.0

8. Appendix II: Error Analysis

Commented [DL29]: 8/10
Include the comparison between the theoretical values and the experimental values.

The experiment will have values that are off from what is theoretically expected as the wet bulb temperature was not accurate. This is in part as the wet bulb membrane fell off during the initial reading which did not allow for water to evaporate correctly and give correct information on temperature. This meant that the readings had to be done in quick succession in order to retain some of the moisture within the metallic couple.

The reading from the hygrometer for humidity and temperature was also questionable as raising the thermocouple without the membrane gave readings with fast evaporated water rather than what would be expected with a large wet membrane. This is definitely the biggest factor that could have resulted in receiving incorrect values that will veer off from the mass and heat transfer coefficients.

Another error in the lab was the air flow constantly needing to be adjusted as the flow was not steady. There were sections in the experiment where the value was recorded and needed to be recorded again as the flow of air was either low or high. There could be an error in some of the values as ± 10% of the value deviation was considered as steady. This was controlled by an individual keeping their attention on the values to adjust accordingly but that raises operator error in consistency.

The top of the reservoir did not have a smooth crevice to allow water to flow evenly in the column. Through visual aid, this was adjusted on the flow of water going in to make the column fully wetted to pass through the thermocouple by a thin layer. This was adjusted by initially turning the flow of water to maximum to increase the chance of water going through the column. Once the column was wetted, it was brought down to a low flow that allowed for trickled-down water to flow in the circumference of the inner tube. This was set as the minimum water flow rate allowed which is 6 SCFM (about 10.19 m^3/h).

However, the water flow rate minimum was not constantly checked to ensure water layer passed through every thermocouple and may have affected the temperature profile in the column. The minimum flow was assumed as the smallest flow that can be allowed and anything higher was considered as being enough to wet the inner column.

9. Appendix III: Sample Calculations

Commented [DL30]: 10/10

To find the mass flowrate in the Gas Phase, G'

Commented [DL31]: Which trial are you showing?

Density of Air: 1275 g/m^3

Flowrate: 0.085 m^3/s

$$G_s' = Density * Flowrate$$

$$G_s' = 1275 * 0.085$$

$$\boldsymbol{G_s' = 1.08E + 02}$$

To find the interface Y', Y'$_i$

Inlet Humidity: 6.10%

Outlet Humidity: 44.50%

Molecular Weight of Air: 29.0 g/mol

Molecular Weight of Water: 18.02 g/mol

$$\mathcal{H} = Inlet\ Humidity + Outlet\ Humidity$$

$$\mathcal{H} = 0.0610 + 0.4450$$

$$\mathcal{H} = 0.5060$$

$$y_i = \frac{\mathcal{H}/M_{Air}}{1/M_{Water} + \mathcal{H}/M_{Air}}$$

$$y_i = \frac{0.506/29.0}{1/18.02 + 0.506/29.0}$$

$$y_i = 0.00313$$

$$Y_i' = \left(\frac{0.00313}{1 - 0.00313}\right)\left(\frac{18.02}{29.0}\right)$$

$$\boldsymbol{Y_i' = 0.00195}$$

The same concept is done for Y_1' and Y_2' but the actual humidity is used instead of the average.

To find the mass transfer coefficient, k_Y

$Y'_i = 0.00195$

$Y'_1 = 0.000235$

$Y'_2 = 0.00172$

Pz = perimeter * height = 0.418m^2

G'_s = 1.08E+02 g/s

$$k_Y = \frac{\ln\left(\frac{Y_i' - Y_1'}{Y_i' - Y_2'}\right) \cdot G_s'}{Pz}$$

$$k_Y = \frac{\ln\left(\frac{0.00195 - 0.000235}{0.00195 - 0.00172}\right) \cdot 1.08E + 02}{0.418}$$

$$\boldsymbol{k_Y = 512\,\frac{g}{m^2 s}}$$

Reynolds number

Air Flowrate: 0.000085 m^3/s

Column Height: 1.09 m

Kinematic Viscosity: 1.57E-05 m^2/s

Cross-sectional Area: 0.0029 m^2

$$Re = \frac{Air\ Flowrate * Column\ Height}{Kinematic\ Viscosity * Cross\ Sectional\ Area}$$

$$Re = \frac{0.000085 * 1.09}{1.57E - 05 * 0.0029}$$

$$Re = \frac{0.000085 * 1.09}{1.57E - 05 * 0.0029}$$

$$\boldsymbol{Re = 2015}$$

To find the heat transfer coefficient, h

$T'_i = 32.25$

$T'_1 = 45$

$T'_2 = 35.5$

$Pz = 0.418 m^2$

$G'_s = 1.08E+02$ g/s

$Cs = Cp_{AIR} + Y'_1 \, Cp_{H2O} = 1.01$ J/g-C

$$h = \frac{\ln\left(\frac{T_i' - T_1'}{T_i' - T_2'}\right) \cdot G_s' \cdot C_s}{Pz}$$

$$h = \frac{\ln\left(\frac{32.25 - 45}{32.25 - 35.5}\right) \cdot 1.08E + 02 \cdot 1.01}{0.418}$$

$$\boldsymbol{h = 356 \frac{J}{m^2 s}}$$

Schmidt Number

Kinematic Viscosity: 1.57E-05 m^2/s

Diffusion Coefficient: 0.00219 m^2/s

$$Sc = \frac{Kinematic\ Viscosity}{Diffusion\ Coefficient}$$

$$Sc = \frac{1.57E - 05}{0.00219}$$

$$\boldsymbol{Sc = 0.0072}$$

Sherwood Number

Reynolds Number: 2015

Schmidt Number: 0.0072

$$Sh = 0.023 \cdot Re^{0.83} \cdot Sc^{0.44}$$

$$Sh = 0.023 \cdot (2015)^{0.83} \cdot (0.0072)^{0.44}$$

$$\boldsymbol{Sh = 1.45}$$

10. Appendix IV: Individual Team Contributions

NAME: Diana Awimrin

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Following Procedure and changing the flowrates of air and changing temperature of steam.
Pre-Lab Editing	10 mins	Typing and Editing
Experimental Objective (Prelab)	5 mins	The overall goal of the lab
Apparatus (prelab)	10mins	The set-up of the instruments and how they work
Materials & Supplies (prelab)	10 mins	List of the supplies needed
Procedure (prelab)	1	Detailed instructions of how to do the lab properly
Project Safety Evaluation (prelab)	30mins	Evaluating how the instrument could bring potential hazards
Final Lab Editing	1	Edited the final Report and proofread
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	2	Discuss the results and relate it with the theory
References (final and/or prelab)	15 mins	Cited the sources that was used
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	Added in information asked from the template PowerPoint
Total	12 hours 20 min	

NAME: Leena Daniel

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Following Procedure and changing the flowrates of air and changing temperature of steam.
Pre-Lab Editing	10 mins	Typing and Editing
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)	10 mins	List of the supplies needed
Procedure (prelab)	1	Detailed instructions of how to do the lab properly
Project Safety Evaluation (prelab)	5mins	Evaluating how the instrument could be potentially hazardous
Final Lab Editing	5 mins	Edited the final Report and proofread
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	4 hours	Data Tables, Calculations for Plots and Results
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	15 mins	Generated Tables used in Results
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	20 mins	Added in information asked from the template PowerPoint
Total	12 hours 20 mins	

NAME: Jay Rawal

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Following Procedure and changing the flowrates of air and changing temperature of steam.
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)	10mins	The set-up of the instruments and how it works
Materials & Supplies (prelab)	10 mins	List of the supplies used to make the solution
Procedure (prelab)	1	Detailed instructions of how to do the lab properly
Project Safety Evaluation (prelab)	30 mins	Evaluating how the instrument could be potentially hazardous
Final Lab Editing	1	Edited the final Report and proofread
Executive Summary (final lab)		
Introduction (Final lab)	1	Re-read lab manual and research online
Theory (Final Lab)	2	Researched Humidification Process
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	30 mins	Added in information asked from the template PowerPoint
Total	12 hours 20 mins	

NAME: Esteban Vazquez

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Following Procedure and changing the flowrates of air and changing temperature of steam.
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)	20 mins	List of the supplies used
Procedure (prelab)	1	Detailed instructions of how to do the lab properly
Project Safety Evaluation (prelab)	30mins	Evaluating how the instrument could be potentially hazardous
Final Lab Editing	1	Edited the final Report and proofread
Executive Summary (final lab)	1	Reviewed all the parts and consolidated it to a summary
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)	15 mins	Cited the sources that was used
Data Tabulation/Graphs (final)		
Error Analysis (final lab)	2	Analyzed the data given and calculated the regression and percent error.
Sample Calculations (final Lab)		
Power Point Presentation	30 mins	Added in information asked from the template PowerPoint
Total	12 hours 20 mins	